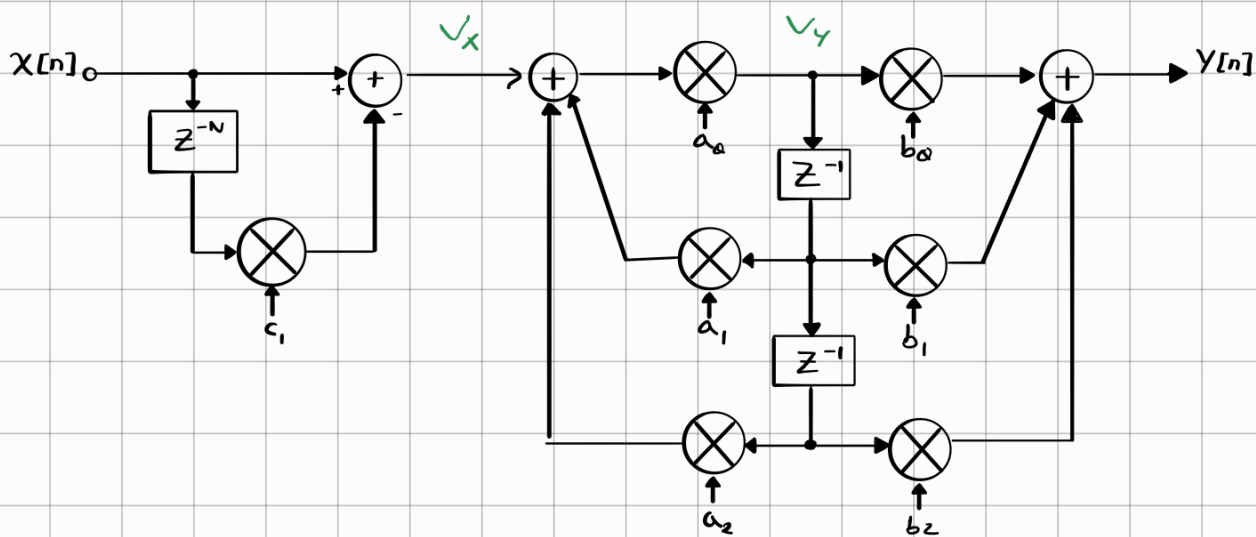




1)

$$Y[n] = (b_0 + b_1 z^{-1} + b_2 z^{-2}) V_y$$

$$V_y = a_0 V_x + a_0 (a_1 z^{-1} + a_2 z^{-2}) V_y$$

$$\hookrightarrow V_y = \frac{a_0}{1 - a_0 a_1 z^{-1} - a_0 a_2 z^{-2}} V_x$$

$$V_x = X[n] (1 - c_1 z^{-N})$$

$$\Rightarrow Y[n] = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}} (1 - c_1 z^{-N}) X[n]$$

$$\Rightarrow H[n] = (1 - c_1 z^{-N}) \cdot \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}$$

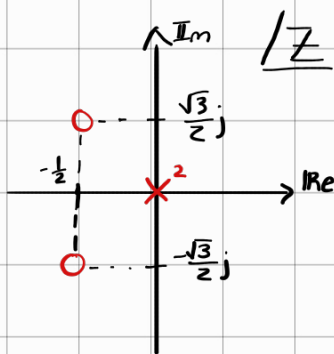
2) $a_0 = 1, a_1 = 1, a_2 = 0, b_0 = \frac{1}{N}, b_1 = 0, b_2 = 0, c_1 = 1, N = \{3, 4, 5\}$

$$H[z] = 1 - c_1 z^{-N} \cdot \frac{\frac{1}{N} + b_1 z^{-1} + b_2 z^{-2}}{1 - z^{-1} - a_2 z^{-2}} = \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}}$$

$$\Rightarrow \boxed{H(z) = \frac{1}{N} \cdot \frac{1 - z^{-N}}{1 - z^{-1}}} \rightarrow H(z) = \frac{1}{N} \cdot \frac{z^N - 1}{z^{N-1}(z - 1)}$$

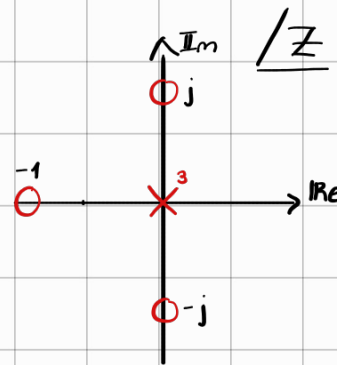
$N=3) H_3(z) = \frac{1}{3} \frac{z^3 + z^2 + z + 1}{z^2}$

$\hookrightarrow H_3(\omega) = \frac{1}{3} \frac{\sin(\frac{3}{2}\omega)}{\sin(\frac{1}{2}\omega)} e^{-j\omega}$



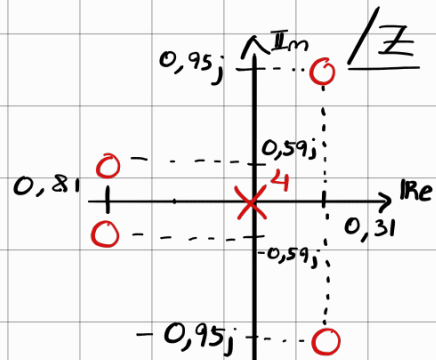
$N=4) H_4(z) = \frac{1}{4} \cdot \frac{z^4 + z^3 + z^2 + z + 1}{z^3}$

$\hookrightarrow H_4(\omega) = \frac{1}{4} \frac{\sin(2\omega)}{\sin(\frac{1}{2}\omega)} e^{-j\frac{3}{2}\omega}$



$N=5) H_5(z) = \frac{1}{5} \cdot \frac{z^5 + z^4 + z^3 + z^2 + z + 1}{z^4}$

$\hookrightarrow H_5(\omega) = \frac{1}{5} \frac{\sin(\frac{5}{2}\omega)}{\sin(\frac{1}{2}\omega)} e^{-j2\omega}$



• Es un Filtro FIR ya que la componente del Integrador (IIR) resulta anulado por el Comb.

• Ventajas: Al tener mayor grado (mayor N) se obtienen más coeficientes y se puede cumplir plantillas más exigente.

$$H(z) = \frac{1}{N} \cdot \frac{z^N - 1}{z^{N-1}(z-1)} = \frac{1}{N} \cdot \frac{z^{N-1} + z^{N-2} + \dots + z^2 + z + 1}{z^{N-1}}$$

$$H_N(z) = \frac{1}{N} \cdot (1 + z^{-1} + z^{-2} + \dots + z^{-(N-2)} + z^{-(N-1)})$$

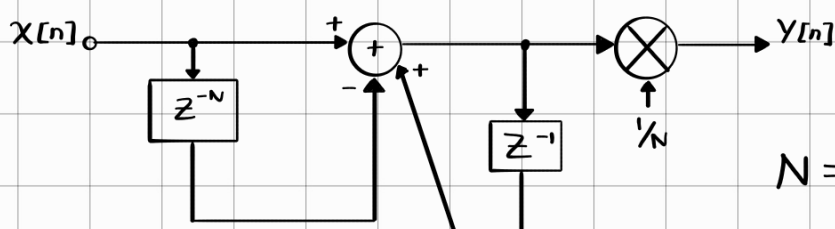
• Para lograr una $h[k] = (1, 1, 1, 1, 1, 1, 1)$ se necesita una $H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6} \Rightarrow N = 7$

$$\hookrightarrow H_7(z) = \frac{1}{7} (1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6})$$

$$\Rightarrow h_7[k] = \left(\frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{1}{7}\right)$$

• Se podría llegar a una $h[k] = (1, 1, 1, 1, 1, 1, 1)$ si se aplica una ganancia de 7 veces externa.

• Diagrama en bloques:



$$N = \{3, 4, 5\}$$

$$3) \quad Y[n] = \frac{1}{2} (x[n] - x[n-2])$$

$$\mathcal{Z}\{Y[n]\} = \mathcal{Z}\left\{\frac{1}{2}(x[n] - x[n-2])\right\}$$

$$Y[z] = \frac{1}{2} (X[z] - X[z] z^{-2})$$

$$Y[z] = \frac{1}{2} (1 - z^{-2}) X[z]$$

$$H[z] = \frac{1}{2} (1 - z^{-2})$$

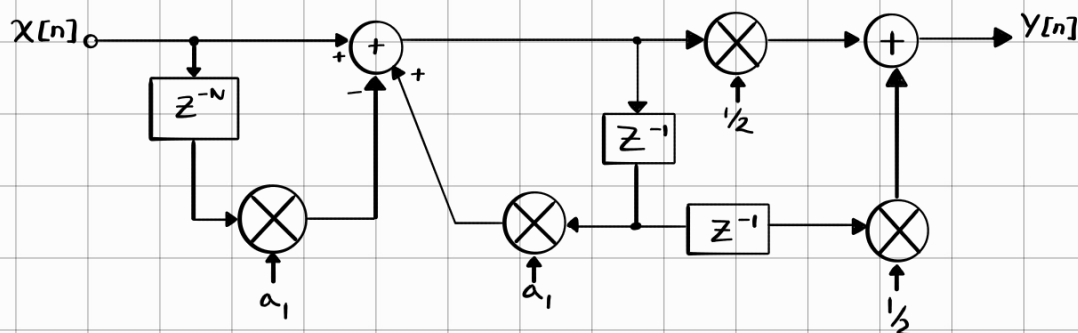
$$H[z] = \frac{1}{2} \frac{z^2 - 1}{z^2} //$$

• Si $H[z] = (1 - c_1 z^{-N}) \cdot \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{a_0 + a_1 z^{-1} + a_2 z^{-2}} / H[z] = \frac{1}{2} (1 - z^{-2})$

$$\Rightarrow \left\{ b_0 = b_2 = \frac{1}{2} \right\}, \left\{ b_1 = 0 \right\}, \left\{ a_0 = 1 \right\}$$

$$\left\{ \begin{array}{l} \text{Si } N=1 : \left\{ a_1 = c_1 \right\}, \left\{ a_2 = 0 \right\} \\ \text{Si } N=2 : \left\{ a_2 = c_1 \right\}, \left\{ a_1 = 0 \right\} \end{array} \right.$$

• Diagrama en bloques:



4) Si $H(z) = (1 - \alpha_1 z^{-1}) \cdot \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{a_0 - a_1 z^{-1} - a_2 z^{-2}} / \left\{ \begin{matrix} a_0 = b_0 = 1 \\ b_1 = -1 \\ a_1 = \beta = 0,9 \end{matrix} \right\}$

$$\Rightarrow H(z) = \frac{1 - z^{-1}}{1 - \beta z^{-1}} = \frac{z - 1}{z - \beta} //$$

• $H(\Omega) = H(z) /_{z=e^{j\Omega}} = \frac{e^{j\Omega} - 1}{e^{j\Omega} - \beta} //$

$\hookrightarrow$ Si $\Omega = 0 \Rightarrow H(0) = 0 \longrightarrow$ Se verifica la eliminación de DC

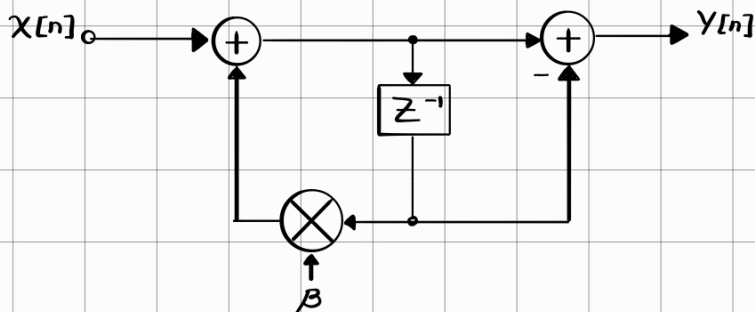
• $H(\Omega) = \frac{e^{j\Omega} - 1}{e^{j\Omega} - \beta} = \frac{e^{j\frac{\Omega}{2}}(e^{j\frac{\Omega}{2}} - e^{-j\frac{\Omega}{2}})}{\cos \Omega + j \sin \Omega - \beta}$

$\hookrightarrow |H(\Omega)| = \frac{2|\sin(\frac{\Omega}{2})|}{\sqrt{(\cos \Omega - \beta)^2 + \sin^2 \Omega}} = \frac{2|\sin(\frac{\Omega}{2})|}{\sqrt{\beta^2 - \beta \cdot 2 \cos \Omega + 1}} //$

$\hookrightarrow 20 \log |H(\Omega = \frac{\pi}{10})| = 20 \log |H(\Omega = \pi)| - 3 \text{ dB}$

$|H(\Omega = \frac{\pi}{10})| = |H(\Omega = \pi)| \times 10^{-\frac{3}{20}} \longrightarrow \boxed{\beta = 0,7271}$

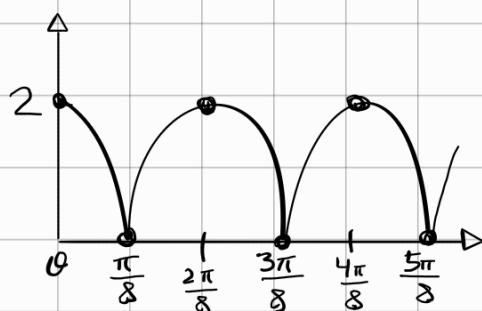
• Diagrama en bloques:



$$5) H[z] = (1 - c_1 z^{-1}) \cdot \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}$$

- $H[z] = 1 + z^{-8} \longrightarrow H[z] = \frac{z^8 + 1}{z^8} //$

- $H(\Omega) = 1 + e^{-j(8\Omega)} = 2 \cos(4\Omega) e^{-j4\Omega} //$



$$\Rightarrow \begin{cases} \text{nulls: } \Omega = \frac{\pi}{8} + \frac{\pi}{4}k \\ \text{peaks: } \Omega = \frac{\pi}{4}k \end{cases}$$

Propongo una $f_{\text{sampling}} = 80 \text{ Hz}$

$$f = \frac{\Omega}{2\pi} f_{\text{sampling}}$$

$f [\text{Hz}]$	$\Omega = 2\pi \frac{f}{f_s}$	Ocorrência
0	0	Peak ($k=0$)
5	$\pi/8$	Null ($k=0$)
10	$\pi/4$	Peak ($k=1$)
15	$3\pi/8$	Null ($k=1$)
20	$\pi/2$	Peak ($k=2$)
$\vdots$	$\vdots$	$\vdots$

- Diagrama en bloques:

